

VARUN K.V

+91 9663089804 | varunkvvarunkv1@gmail.com | github.com/varun1-code | linkedin.com/in/varun-kv-bunny | Bangalore, Karnataka

PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience building LLM-powered applications, full-stack web platforms, and machine learning systems. Skilled in prompt engineering and structured LLM output design with practical experience in hallucination control, quote-grounded extraction, and deploying LLM applications via Streamlit and Docker. Backend experience with Python (FastAPI), Ruby on Rails, REST and GraphQL APIs, and third-party vendor integrations. Comfortable communicating across technical and non-technical stakeholders. Eager to learn agentic AI workflows and contribute to client-facing AI implementations.

TECHNICAL SKILLS

LLMs & Prompt Engineering: Prompt Design, Structured JSON Output, Hallucination Control, Quote-Grounded Extraction, OpenAI-compatible LLM APIs, Claude (Anthropic), HuggingFace Transformers

AI / ML: PyTorch, TensorFlow, Scikit-learn, CNNs, Computer Vision, Transfer Learning, Model Evaluation

Backend & APIs: Python, FastAPI, Flask, Ruby on Rails, REST APIs, GraphQL, Sidekiq, Webhooks

Frontend: React, Vite, Tailwind CSS, Streamlit

Data & Document Processing: SQL, SQLite, Pandas, NumPy, Redis, Power BI, PyMuPDF, python-docx, Tesseract OCR, ReportLab

DevOps & Tools: Docker, Git, GitHub, Pytest, Vitest, Postman, Linux

PROJECTS

Earnings Call Research Tool / *Streamlit · Python · LLM API · Docker · OCR*

Personal / *Structured LLM Output · Prompt Engineering · Hallucination Control*

- Built a structured LLM-powered research tool that analyses earnings call transcripts and management commentary, producing analyst-ready outputs: management tone, confidence level, key positives, concerns, forward guidance (revenue / margin / capex), capacity utilization, and growth initiatives.
- Engineered prompts to enforce strict transcript-grounding — every extracted point must include a supporting direct quote, and missing sections return *null* instead of fabricated content.
- Built a multi-format ingestion pipeline (PDF, TXT, DOCX) using PyMuPDF and python-docx, with Tesseract + Poppler OCR fallback for scanned documents.
- Integrated an OpenAI-compatible LLM endpoint (NVIDIA-hosted) for structured analysis; exported analyst-ready results to JSON and PDF using ReportLab.
- Containerised and deployed the application as a Docker service accessible via a public URL, with OCR dependencies bundled inside the container.

Trupti Yoga & Nature Cure — Full-Stack Membership Web Application / *React 19 · Vite · Tailwind CSS v4 · FastAPI · SQLite · REST API*

Live: trupti-yoga-website.vercel.app

- Built a production-ready membership platform for a live online yoga business — React frontend + FastAPI backend with SQLite database — handling user registration, UPI payment verification, admin approval workflow, and real-time Google Meet session access.
- Implemented JWT-style authentication with SHA-256 password hashing and session token management.
- Built a manual payment verification workflow where users submit UPI transaction IDs and admins approve/reject through a custom admin panel.
- Developed a real-time session status system using IST timezone detection to show live / upcoming / ended state for 3 daily sessions, auto-refreshing every 30 seconds.

- Wrote **53 unit tests** — 30 backend (pytest) and 23 frontend (Vitest) covering auth, payment, session timing, and timezone conversion.

Skin Disease Detection — Deep Learning (Computer Vision) / *PyTorch · CNN · Transfer Learning*

Academic / End-to-end ML pipeline · HAM10000 + Derm7pt datasets

- Built and trained CNN-based classification models on HAM10000 and Derm7pt dermatology datasets (10K+ labeled images, 7 disease classes).
- Applied data preprocessing (normalization, augmentation) and handled class imbalance using weighted sampling; used transfer learning to improve generalization on limited data.
- Constructed an end-to-end training and real-time inference pipeline with accuracy, F1-score, and confusion-matrix tracking.

WORK EXPERIENCE

Backend Developer Intern | *Learnyst (EdTech Platform)*

Bangalore, Karnataka | Feb 2026 – May 2026

- Shipped production backend features at an EdTech startup using **Ruby on Rails, Sidekiq, Redis, GraphQL, and Webhooks**.
- Integrated multiple third-party vendor APIs behind a clean abstraction using the **Strategy design pattern**, enabling providers to be swapped without touching business logic — a pattern that maps directly to integrating multiple LLM providers.
- Built admin-facing reporting flows and worked across REST and GraphQL APIs, background job processing, and structured-log debugging.
- Collaborated via Git workflows (branching, PRs, code reviews) in an agile environment, communicating with engineers and stakeholders throughout the feature lifecycle.

EDUCATION

Maharaja Institute of Technology, Mysore — *Completed June 2026*

B.E. in Computer Science • CGPA: 8.0

BGS PU College — *Completed July 2021* • Pre-University (PUC) • 94.36%

CKS English School, Hassan — *Completed April 2019* • SSLC • 91%

CERTIFICATIONS

- **Artificial Intelligence Fundamentals** — IBM SkillsBuild (Nov 2025)
- **AI & Data Science Applications** — TejasPro Trainings (2025)
- **Python Full Stack Course** — Pentagon Space (2026)
- **Backend Developer Intern** — Learnyst (2026)